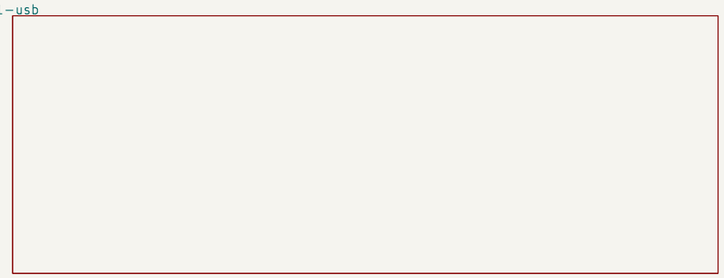
OPENKEYS R0.4 – ONE USB-C + NFC / WHITE 16x2 LCD
Fingerprint mandatory; PIN additional only
PCB candidate; qualification required before secret provisioning

01-usb



File: 01-usb.kicad_sch

02-power



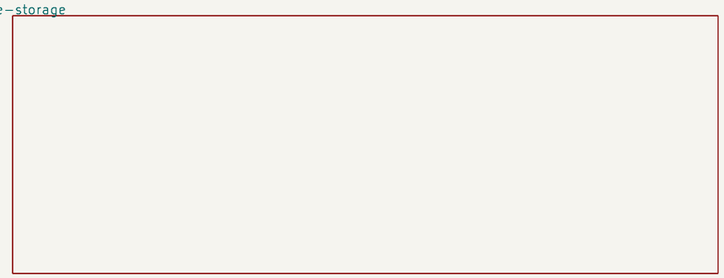
File: 02-power.kicad_sch

03-mcu



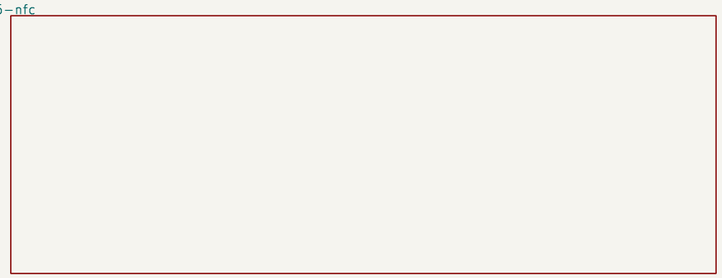
File: 03-mcu.kicad_sch

04-se-storage



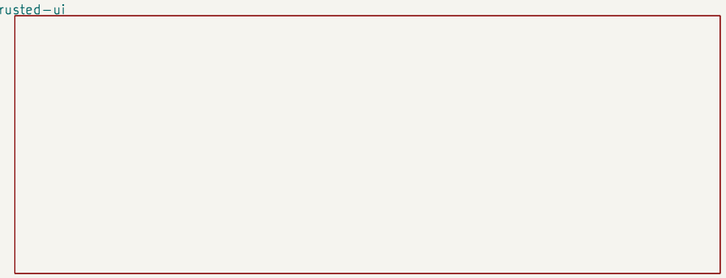
File: 04-se-storage.kicad_sch

05-nfc



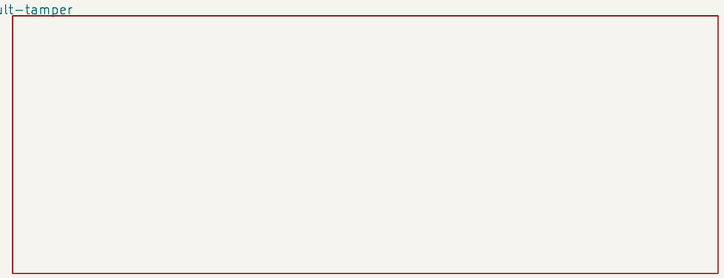
File: 05-nfc.kicad_sch

06-trusted-ui



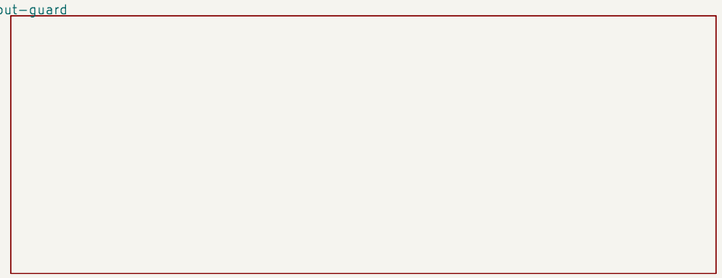
File: 06-trusted-ui.kicad_sch

07-fault-tamper



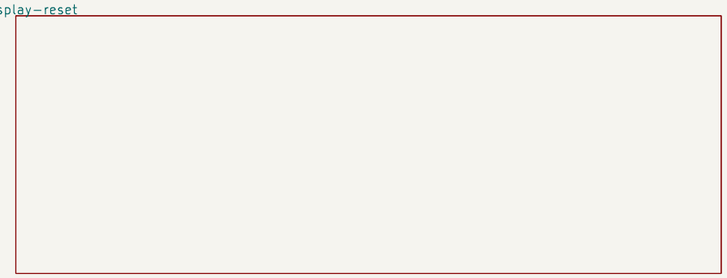
File: 07-fault-tamper.kicad_sch

08-input-guard



File: 08-input-guard.kicad_sch

09-display-reset



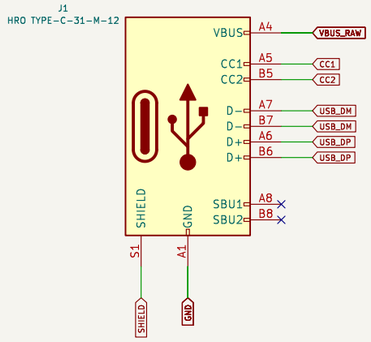
File: 09-display-reset.kicad_sch

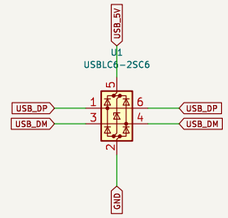
Jay-Void-BA
Sheet: /
File: OpenKeys.kicad_sch

Title: OpenKeys R0.4 – Engineering review

Size: A3	Date: 2026-11-08	**Rev: 0.4**
KiCad E.D.A. 9.0.2		Id: 1/10

One USB-C receptacle — USB 2.0 FS data and 5 V charging

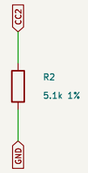


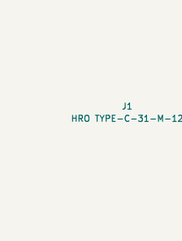






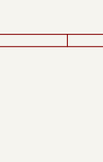












5.1k Rd on each CC pin: sink only. No PD negotiation, no 9 V request.
Apple 20 W adapter uses 5 V through a standards-compliant USB-C cable.
100mA hardware input limit: 5 V x 0.1 A = 0.50 W nominal input budget.
Input guard U14 is independent of MCU: see sheet08. Analog timing/thermal protection not yet qualified.

ENGINEERING BOARD CANDIDATE — qualify before use

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0.4 — 01-usb

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 2/10

Low-rate charging, protected 500 mAh pack and 3.3 V supply

#FLG3
PWR_FLAG

VBUS_FUSED

VBUS_RAW

F1
250mA hold – qualify I2t

VBUS_FUSED

U2
BQ25185DLHR

SYS	1	SYS	
BAT	2	BAT	
CHG_STAT2_N	3	STAT2	
CHARGE_DISABLE	4	CE_N	
GND	5	GND	
BAT_NTC	6	TS_MR	
	7	ILIM_VSET	CHG_ILIM_VSET
	8	ISET	CHG_ISET
	9	STAT1	CHG_STAT1_N
	10	IN	USB_5V
	11	EP	GND

J2
Adafruit 1578 protected 1S 500mAh

BAT	1
GND	2

CHG_ILIM_VSET

R6
24k 1%

GND

CHG_ISET

R7
3.00k 1%

GND

SYS

R8
100k

CHARGE_DISABLE

CHARGE_DISABLE

CHARGE_ENABLE 1

Q2
2N7002

2

3

GND

CHARGE_ENABLE

R9
100k

GND

J6
103AT-2 external battery thermistor

BAT_NTC	1
GND	2

SW_L1

L1
1.5uH Isat>=1.5A

SW_L2

SYS

R10
0R

REG_VINA

#FLG4
PWR_FLAG

REG_VINA

U3
TPS63031DSK (3.3 V)

SYS	5	VIN	
REG_VINA	8	VINA	
REG_EN	6	EN	
GND	7	PS/SYNC	
SW_L1	4	L1	
SW_L2	2	L2	
GND	9	GND	
GND	3	PGND	
	1	VOUT	+3V3
	10	FB	+3V3

USB_5V

C2
1uF 10V

GND

BAT

C3
10uF 6.3V

GND

SYS

C4
10uF 10V

GND

REG_EN 2

SW1
OFF / ON

1

3

SYS

GND

SYS

C5
10uF 10V

GND

REG_VINA

C6
100nF

GND

+3V3

C7
10uF 6.3V

GND

+3V3

C8
10uF 6.3V

GND

Fixed100mA input; charge target100mA=0.2C for500mAh.
Input DPM gives system load priority, so cell charge can be slower.
CE defaults disabled. Public charging service never unlocks secrets.
Suspend: stop charge/backlight; qualify total VBUS current <=2.5mA.
Real pack PCM + attached thermistor required. No fast charge.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0.4 – 02-power

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 3/10

STM32H573RIT6 — STiRoT and TrustZone protected I/O

C9
100nF

C10
100nF

C11
100nF

C12
100nF

C13
100nF

C14
4.7uF

C15
2.2uF low ESR

C16
2.2uF low ESR

C17
100nF

C18
1uF

U4
STM32H573RIT6

FB1
600R@100MHz

#FLG6
PWR_FLAG

R11
10k

C19
10nF — qualify reset timing

R12
10k

C20
9pF C0G — tune

C21
9pF C0G — tune

Y1
ABS07-32768KHZ-6-T

J3
Internal SWD — DNP production

STiRoT must verify protected firmware. TrustZone/GTZC must deny nonsecure access to FP, SE, LCD, buttons, timers, DMA and key memory.
No raw cryptography or arbitrary display commands at secure gateway.
SE_EN or a sensor MATCH wire is NOT fingerprint authentication.
Separate2.2uF on VCAP1/2; no external load on VCAP.

ENGINEERING BOARD CANDIDATE — qualify before use

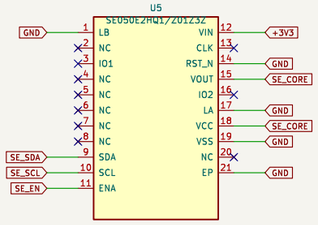
Jay-Void-BA

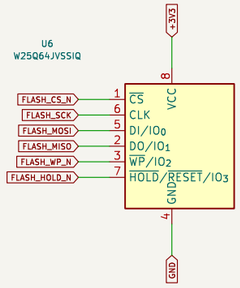
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

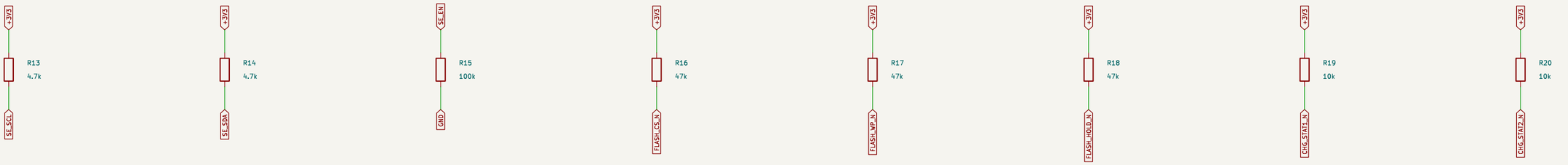
Title: OpenKeys R0.4 — 03-mcu

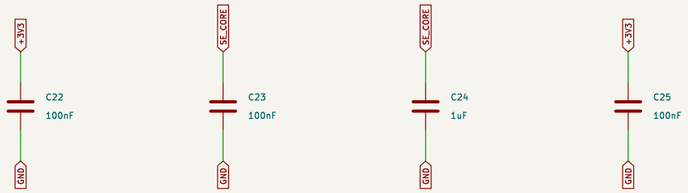
Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 4/10

Secure element and encrypted storage — no OTP / wall-clock RTC









Each user security operation: fresh authenticated device FP + confirmation.
Sensitive administration also needs device PIN. PIN never replaces FP.
Precharge attempts durably: uncertain writes deny. AEAD does not alone prevent rollback.
SE certification does not certify MCU, PCB, biometric path or complete product.

ENGINEERING BOARD CANDIDATE — qualify before use

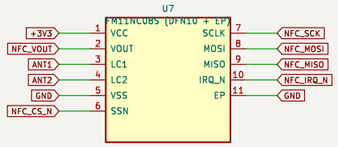
Jay-Void-BA

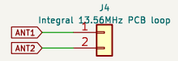
Sheet: /04-se-storage/
File: 04-se-storage.kicad_sch

Title: OpenKeys R0.4 — 04-se-storage

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 5/10

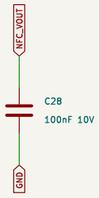
NFC Type4 card interface — battery-powered trusted fingerprint UI

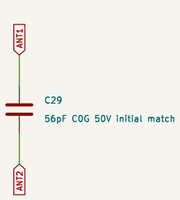
















NFC VOUT isolated from SYS/BAT/+3V3. No assumed RF-powered full device.
FP/LCD/CTAP NFC require charged battery and device ON.
RF region: tuned coil only, no cell, metal shield or ground planes.
Mobile management needs authenticated APDUs and native OS integration.
Phone biometrics or a tap never replace device fingerprint verification.

ENGINEERING BOARD CANDIDATE — qualify before use

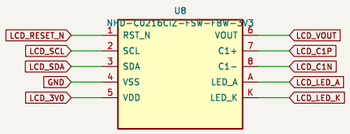
Jay-Void-BA

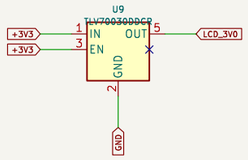
Sheet: /05-nfc/
File: 05-nfc.kicad_sch

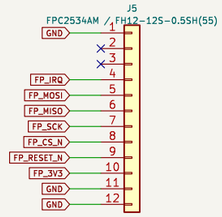
Title: OpenKeys R0.4 — 05-nfc

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 6/10

White-backlit16x2 LCD, authenticated fingerprint and contextual buttons











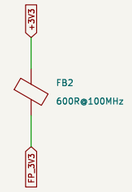


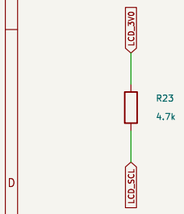




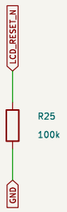




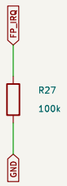






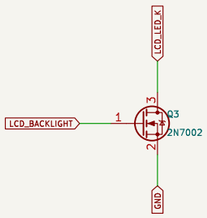


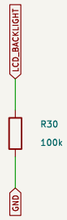




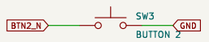


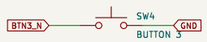






















ENGLISH16x2; page full identity before confirmation.
PIN: short1..4=symbol; hold4=erase; hold2=cancel.
Menu:1=prev,2=next,3=confirm,4=cancel. No host injection.
Backlight about10mA via33R; default OFF.

ENGINEERING BOARD CANDIDATE — qualify before use

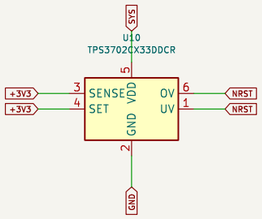
Jay-Void-BA

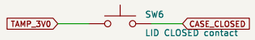
Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

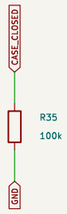
Title: OpenKeys R0.4 — 06-trusted-ui

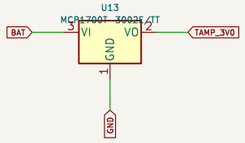
Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 7/10

Independent reset, battery–backed case tamper and SE interlock



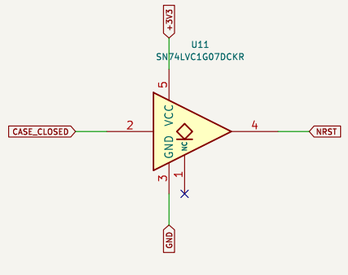


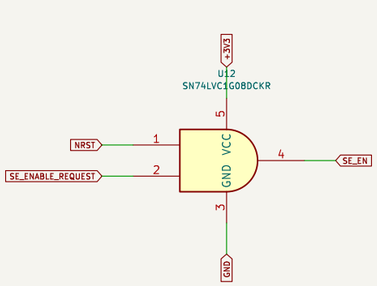




















Reset invalidates volatile grants; reboot must verify firmware and require fresh FP.
PC13=case input; PC1/TAMP_IN3 observes reset/fault line.
Internal BOR/PVD, clock checks and persistent precharged attempts also required.
Production debug lifecycle closed; no populated external debug connector.
VBAT+LSE must retain tamper with main OFF; full backup loss must lock, not initialize.
Measure reset/ENA latency, leakage and injection before security release.

ENGINEERING BOARD CANDIDATE – qualify before use

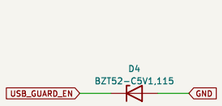
Jay–Void–BA

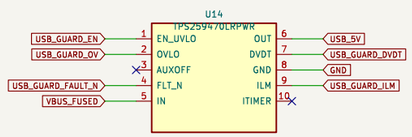
Sheet: /07–fault–tamper/
File: 07–fault–tamper.kicad_sch

Title: OpenKeys R0.4 – 07–fault–tamper

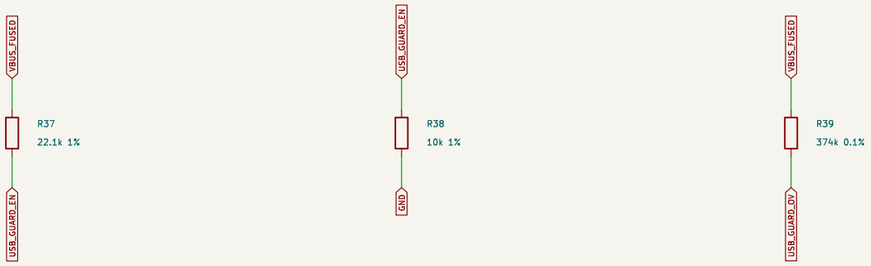
Size: A3	Date: 2026–11–08	**Rev: 0.4–board–review**
KiCad E.D.A. 9.0.2		Id: 8/10

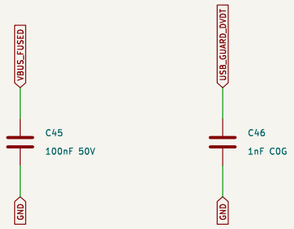
Independent VBUS overvoltage cutoff, reverse blocking and limited inrush











OVLO nominal1.2V x(1+374k/100k)=5.688V. Component/leakage corners in protection-calculations.json.
UVLO divider22.1k/10k: about3.85V nominal. EN is separately clamped, never raw VBUS.
6.49k ILM: about0.51A protection threshold; BQ charger still sets100mA input budget.
1nF DVDT: about2V/ms, about2.5ms ramp to5V; measure actual inrush with complete load.
F1 -> input TVS -> eFuse -> output TVS/ESD reference/charger/sense.
Independent cutoff is a circuit provision; response overshoot, CC/data faults and thermal/surge tests remain BLOCKED.

ENGINEERING BOARD CANDIDATE - qualify before use

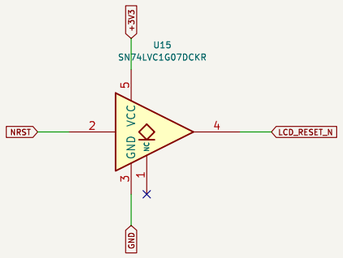
Jay-Void-BA

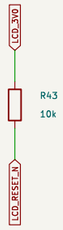
Sheet: /08-input-guard/
File: 08-input-guard.kicad_sch

Title: OpenKeys R0.4 - 08-input-guard

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
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Independent physical LCD clearing and level-safe reset







NRST asserts from supervisor, case-open buffer or MCU reset.
U15 drives LCD_RESET_N low without waiting for firmware startup.
PC3 can also pull low on lock, expiry, cancel or display fault.
PC3 open-drain: SCL/SDA open-drain on3.0V pull-ups.
No host raw-display command. Hardware reset is not a substitute for fingerprint gating of display content and authenticated transactions.
Validate reset minimum width and complete physical LCD clearing.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /09-display-reset/
File: 09-display-reset.kicad_sch

Title: OpenKeys R0.4 – 09-display-reset

Size: A3	Date: 2026-11-08	**Rev: 0.4-board-review**
KiCad E.D.A. 9.0.2		Id: 10/10